Devi Prasanth Karumanchi

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Professional Summary

Computer science MS Candidate (University of North Texas,2026) who designs and deploys end to end machine learning applications, shipped two production Machine Learning web applications on Render: a Hybrid movie recommender (TF-IDF + collaborative filtering served through a REST API) and a customer-segmentation engine that processed 540K+ transactions to reveal that 16% of customers that drive 65% of revenue. Skilled in Python, Scikit-Learn, and the Pandas/NumPy stack, with hands-on work in regression, K-Means Clustering, feature engineering, and model deployment

Skills

Machine Learning: Supervised & Unsupervised Learning**,** Regression, Classification, K-Means Clustering, Recommendation Systems, Collaborative & Content-Based Filtering, Feature Engineering, PCA/Dimensionality Reduction, Train/Test Split, Model Evaluation
Deep Learning: Neural Networks, Convolutional Neural Networks, TensorFlow/Keras, Pytorch
Data Analysis: Pandas, NumPy, Matplotlib, Data Cleaning, Exploratory Data Analysis (EDA), Data visualization, RFM analysis,
Programming: Python, R, SQL, C, HTML/CSS
Cloud & Deployment: AWS (S3, EC2), Flask, REST API, Model Deployment (Render)
Tools: Jupyter Notebook, Git & GitHub, VS code

Projects

1. Movie Recommendation System (Hybrid ML Web App) | *Python, Git, Scikit-Learn, Flask, REST API*

- Engineered a full-Stack Hybrid movie recommendation system in Flask, Integrating ML models with a responsive web interface
- Boosted predictive performance, Increasing R^2 nearly 10-fold (0.03 → 0.31) through feature engineering on user/movie average ratings, and cut error to MAE 0.73 and RMSE 0.92.
- Combined content-based filtering (TF-IDF + cosine similarity) with collaborative filtering (user rating patterns) to generate personalized recommendations.
- Designed REST APIs to process user input and serve the real-time recommendations, then deployed the application on render for public access

2. E-Commerce Customer Segmentation System (ML Web App*) | Python, Scikit-Learn, K-Means, PCA, Flask*

- Built and end to end segmentation pipeline over 540K+ e-commerce transactions, engineering RFM (Recency, Frequency, Monetary) features from raw, cleaned data.
- Trained a K-Means clustering model to group 4338 customers into 4 behavioral segments, validating cluster count with the elbow method and silhouette score
- Applied log transformation, standardization, and PCA to compress features to 2D for visualization while retaining 94 % of variance
- Uncovered that 16% of customers drive 65% of revenue, translating segments into targeted marketing recommendations and deployed the model as a live flask web application for real-time segment prediction

Education

Master of Science — Computer Science and Engineering
University of North Texas, Denton, TX 2024-2026

Bachelor of Technology — Computer Science and Engineering (Specialization in Gaming and Graphics)
Chandigarh University, Mohali, India 2019-2023